

ADVANCED CASE STUDY

Multi-Branch Retail & Service Management System

Database Design, ERD, Relational Mapping & SQL Server Implementation

Company Name: Nova Retail Group

1. Scenario

Nova Retail Group operates multiple branches and sells products supplied by different suppliers. The company wants to replace its current disconnected files with a centralized relational database hosted on **SQL Server**.

The system must support customers, employees, departments, branches, products, suppliers, orders, order details, and payments. The database must also enforce data integrity and support analytical SQL queries.

2. Business Requirements

1. **Customer Identifiers:** Each customer must have a unique CustomerID. Different customers may have the same name. Phone numbers and email addresses may be missing.
2. **Employment Allocation:** Each employee works in *exactly one* department and *one* branch. A department and a branch can each contain multiple employees.
3. **Hierarchy & Supervision:** An employee may supervise other employees. A supervisor is also stored as an employee in the same system. Some employees, such as senior managers, may have no supervisor.
4. **Product Categorization:** Each product belongs to *one* category, while each category can contain many products.
5. **Multi-Sourcing Logistics:** A product can be supplied by multiple suppliers, and each supplier can supply multiple products.
6. **Agreed Purchase Pricing:** The purchase price of the same product may differ from one supplier to another. Therefore, the supplier-product relationship must store the agreed purchase price.
7. **Customer Purchases:** Each customer may place multiple orders, but every order belongs to *one* customer.
8. **Order Processing:** Every order is processed by *one* employee and belongs to *one* branch.
9. **Order Items:** An order may contain multiple products, and the same product may appear in many different orders.
10. **Historical Unit Pricing:** The system must preserve the unit selling price at the time of the order, even if the product's current selling price changes later.

11. **Order Quantity Integrity:** The quantity of an ordered product must always be greater than zero (Quantity > 0).
12. **Financial Value Constraints:** Unit prices and payment amounts must not be negative (Amount >= 0).
13. **Flexible Payment Structure:** An order may have no payment yet, one payment, or multiple payments. For example, a customer may pay part of an order using **Cash** and the remaining amount using **Card**.
14. **Allowed Payment Methods:** Valid payment methods are restricted strictly to: Cash, Card, Wallet, Bank Transfer.
15. **Order Lifecycle Statuses:** Order status must be restricted to: Pending, Completed, Cancelled, Returned.
16. **Orphan Prevention:** A product that has already been referenced in an order must not be deleted if this would create orphaned order-detail records.
17. **Data Quality & Edge Cases:** Historical customer data contains quality issues:
 - Names with leading/trailing spaces.
 - Inconsistent capitalization.
 - Phone numbers containing spaces.
 - Missing emails stored either as NULL or 'N/A'.
18. **Data Integrity Enforcement:** The database must enforce **Entity Integrity**, **Referential Integrity**, and **Domain Integrity** using appropriate keys and constraints.

3. Required Tasks

Task 1 — Entity Relationship Diagram (ERD)

Design a complete **Entity Relationship Diagram (ERD)** for Nova Retail Group.

Your ERD must clearly show:

- Entities & Attributes
- Primary Keys & Foreign Keys
- Cardinalities & Structural Constraints
- Optional and Mandatory Relationships
- One-to-Many (1: N) Relationships
- Many-to-Many (M: N) Relationships
- Recursive / Self-Referencing Relationships

Design Directive: You must determine the appropriate design for relationships such as **Products and Suppliers**, **Orders and Products**, and **Employees and Supervisors**. Do

not store a Many-to-Many ($M:N$) relationship directly without resolving it appropriately into an associative entity.

Task 2 — Relational Mapping

Convert the ERD into a relational schema.

For every table, specify:

- Table Name
- Column Names & Data Types
- Primary Keys (PK)
- Foreign Keys (FK)
- NOT NULL Constraints
- UNIQUE Constraints (where appropriate)
- CHECK Constraints (where appropriate)